Diem’s size hypothesis and an actual small-field control

Lean checks the arithmetic last step of Diem2011 Proposition4.11: for natural n,q,N, the following implication holds. The geometric estimate is an explicit premise. [pcc:unverified]

$$n \geq 2, \quad 2^{7n} \leq q, \quad N \geq q + 1 - n2^{n+2}(\sqrt{q} + 1) \quad \Longrightarrow \quad q + 1 \leq 2N.$$

It also constructs an actual nonsingular F4 curve with one rational point and an actual point/chart factor base of cardinality0. This refutes the unrestricted strengthening; the source’s size/admissibility hypotheses remain explicit. [pcc:unverified]

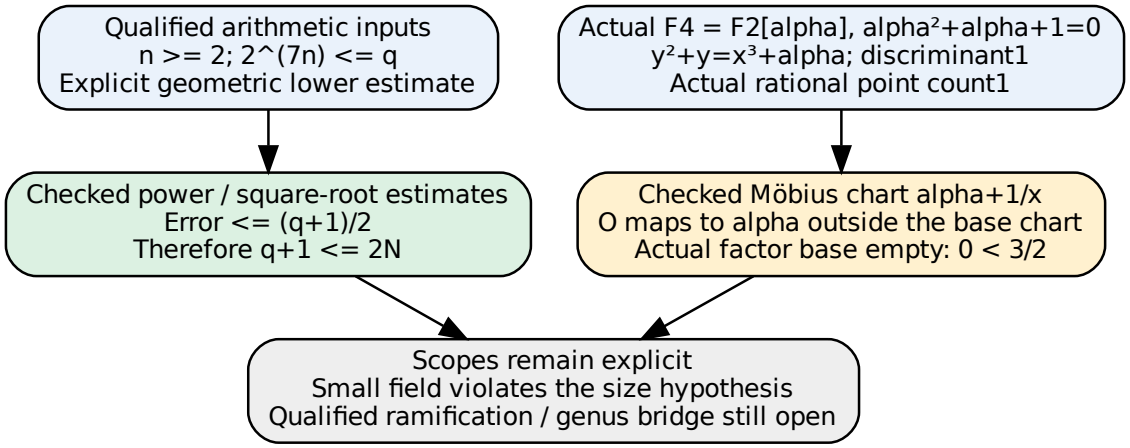


Figure 1: Checked arithmetic and actual model, with distinct scopes. [pcc:unverified]

What is proved, and what remains qualified

For n≥2, $n \le 2^{n-1}$ gives $(n2^{n+4})^2 \le 2^{4n+6} \le 2^{7n} \le q$. With $s = \sqrt{q}$ and $a = n2^{n+2}$, we have $s \ge 4a$ and $s \ge 1$. Thus $a(s+1) \le 2as \le q/2$, giving the conditional half-size bound. An arithmetic-only control n2/q4/N0 directly disproves inference from the geometric estimate alone after omitting the size hypothesis. That N0 is not asserted to be an actual curve count. [pcc:unverified]

The actual model uses $F = \mathbf{F}_2[t]/(t^2+t+1)$ with $\alpha = [t]$. Lean checks irreducibility, degree2 and cardinality4. The curve $y^2 + y = x^3 + \alpha$ has discriminant1. Fourth powers equal the element, so $y^2 + y$ and x^3 are fixed by squaring; the curve equation would force $\alpha^2 = \alpha$, a contradiction. Only O remains: actual point count1. The homography $\varphi = \alpha + 1/x$ has a checked inverse and sends O to alpha outside the F2 projective chart, so the actual factor base is empty. Its half-size inequality $3 \le 2 \cdot 0$ is false. [pcc:unverified]

Main binding: `DiemSmallFieldAudit.diem_domain_audit_certificate`. Proof map: `../../DIEM_DOMAIN.md`. Exact hypotheses, source hashes, gate and actual packet standing are recorded separately in `../../diem-domain-sources.json` and the verification record. Whole census gate535passes, zero failures/zero trusted-native exceptions. Local source commit `6e8973a3` precedes packet `PCC-20261110-dda065`; actual local verdict and signed standing are recorded separately. Lean4.35.0/Mathlib56a00585; standard axioms propext, Classical.choice, Quot.sound only. No hole, custom axiom, native-decision exception, elaboration option or linter waiver. Failed attempts are retained; fresh lean4checker is unavailable. [pcc:unverified]

Original Diem2011 Condition2.7 and Props4.9–4.11 were read; Prop4.11 and proof visually checked on PDFpage22/printed95. DOI 10.1112/S0010437X10005075, PDF SHA25674902b58...934a58a. Reviewed Mathlib quotient-field/power-basis/cardinality, characteristic transport, real sqrt and actual elliptic point constructors/coordinates. KB/Conductor unavailable; bounded primary/repository/Mathlib fallback, no novelty or blanket source promotion. [pcc:unverified]

The source assumes n≥2 and $\log_2 q \ge 7n$, as well as Condition2.7. The actual control has q2/n2 and violates that size condition. Scheme cover degree/ramification, admissibility identification and the geometric Hasse–Weil/genus estimate are not formalized here. The qualified source proposition is not refuted. No native experiment, timing or paid compute. Signed standing remains unverified pending a trusted signed verdict. [pcc:unverified]